MILITARY SPECIFICATION

LINER ASSEMBLY OUTER, FOR CARTRIDGE 106MM, M346A1

1. SCOPE

1.1 This specification covers the liner assembly for Cartridge, 106MM, M346A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-P-10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-1167 — Ammunition Data Card.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8837341 — Outer Liner Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-8837341 — Index of Inspection Equipment Lists.

FSC 1315

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. Parts shall comply with all requirements specified on Drawing (dwg.) 8837341, and with all requirements specified in applicable specifications.

3.3. Workmanship. All manufacturing and processing operations shall be performed in a thorough, workmanlike manner. All parts shall be free of cuts, holes, tears, cracks, splits, excessive flash, dirt, grease or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor or when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certficates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to

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the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 bodies and 25 bases, and 25 assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All mold cavities shall be proportionally represented, molded components shall be identified as to mold and cavity numbers. All materials shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this document and any or all requirements of the applicable drawings.

4.2.3 *Rejections.* If any assembly or component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 *Lot formation.* A lot shall consist of liners produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot number. Lots shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235, when applicable).

4.3.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity,

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AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Body, outer liner (see dwg. 8837342 covering a detail of dwg. 8837341).

Categories	*Defects*	*Method of Inspection*	*Code No. (see 6.2)*
Crticial: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Total length	Gage	01001
202.	Thickness	Gage	01002
203.	Evidence of poor workmanship (see 3.3)	Visual	01003

4.3.2.2 Base, outer liner (see dwg. 8837343 covering a detail of dwg. 8837341)

Categories	*Defects*	*Method of Inspection*	*Code No.*
Crtical: None defined			
Major: None defined			
Minor:	AQL 4.0 percent		
201.	Total length	Gage	02001
202.	Outside diameter of primer boss	Gage	02002
203.	Inside diameter of primer boss	Gage	02003
204.	Length of primer hole	Gage	02004
205.	Depth to bottom of primer hole recess shoulder	Gage	02005
206.	Width at bottom of primer hole recess shoulder	Gage	02006
207.	Angle above primer hole shoulder	Gage	02007
208.	Outside diameter	Gage	02008
209.	Diameter of recess for body	Gage	02009
210.	Wall thickness of recess for body	Gage	02010
211.	Outside profile	Gage	02011
212.	Flashing in primer hole	Visual	02012
213.	Evidence of poor workmanship (see 3.3)	Visual	02013

4.3.2.3 Liner assembly (dwg. 8837341)

Categories	*Defects*	*Method of Inspection*	*Code No.*
Crtical: None defined			
Major:	AQL 0.40 percent		
101.	Heat seal incomplete or incorrect	Visual-Manual	03001
102.	Damage affecting functioning	Visual	03002
Minor:	AQL 1.5 percent		
201.	Total length	Gage	03003
202.	Overlap of component parts complete or incorrect	Gage-Visual	03004
203.	Evidence of poor workmanship (see 3.3)	Visual	03005

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4.3.3 *Inspection Equipment.* Index of Inspection Equipment Lists Number IEL-8837341 identifies the inspection equipment required to perform the examinations and tesst prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.3.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equpiment, construction, survey and whenever inspection equipment drawings are needed.

4.3.3.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with MIL-I-45607.

4.3.3.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing. The packaging and packing shall be in accordance with MIL-P-10025.

5.2 Marking. Marking shall be in accordance with MIL-P-10025.

5.3 Data card. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

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CDAF – Contractor's design responsibility on Army format in accordance with MIL-D-45608.

CDCF – Contractor's design responsibility on contractor format.

AD – Army design.

ADMU – Army design, mandatory for use.

CE – Commercial equipment.

SCD – Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory dseigns and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be

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suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35mm microfilm process. Contractor shall submit one copy of the final approved design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples. Instructions as to the location of evaluation initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND, Dover, New Jersey.

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